

Training and comparing five models

Five modeling approaches were trained and tuned, each searched over a grid of settings using 5-fold cross-validation on the training data, then compared on the untouched validation

Model	What was tuned	Best settings found	Why it was tried
Linear Regression	No tuning (baseline)	N/A	Simplest possible baseline - a straight-line relationship between features and claim
Polynomial Regression	Degree (2 vs. 3)	Degree 2 (degree 3 overfit - CV R^2 dropped to 0.27)	Tests whether a curved, rather than straight, relationship fits better
Random Forest	Tree count, depth, split rules	200 trees, max depth 10, min samples split 5, min samples leaf 2	Can capture interactions and non-linear effects automatically
Support Vector Regression (SVR)	Kernel, regularization (C), margin (epsilon)	Linear kernel, C=10, epsilon=0.5	A different mathematical approach, useful as a sanity check against the others
XGBoost	Tree count, depth, learning rate, subsampling	100 trees, max depth 3, learning rate 0.05, subsample 1.0	Often the strongest performer on structured, tabular data like this

The validation leaderboard - where the winner was picked

Model	R^2	MAE	RMSE
XGBoost	0.79	\$3,881.65	\$5,452.43
Random Forest	0.78	\$3,916.77	\$5,556.01
Linear Regression	0.75	\$4,356.43	\$5,984.24
Polynomial Regression (deg. 2)	0.73	\$4,413.14	\$6,148.15
Support Vector Regression	0.67	\$5,132.24	\$6,810.71

XGBoost had the best validation R^2 (0.79) and the lowest average error, and its 5-fold cross-validation score (0.80) was consistent with its validation performance - a good sign it isn't just a lucky split. XGBoost was selected as the production model.

The one-time test-set result

With the winner already chosen on validation data, the test set - held back and untouched until this point - was used exactly once to report final performance:

0.8337 Test R^2	\$3,751.11 Test MAE	\$4,896.14 Test RMSE
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The model explains about 83% of why claim amounts differ from one policyholder to the next, and its typical prediction is within roughly \$3,751 of the actual claim. As documented and in the Visualization &